MORONI: Polygon-Guided Cellular Textures via Roots-of-Unity Polygons and Phase-Modulated Boundaries

Phani Srikar
Independent Researcher

ABSTRACT

Voronoi and Worley noise are foundational cellular primitives in real-time graphics, but offer limited art direction. Under nearest-site assignment, every cell boundary is constrained to be a perpendicular bisector, preventing artists from directly prescribing target polygon silhouettes, uniform grout widths, or non-linear crack-like seams.

This paper presents **MORONI**, a controllable cellular texture technique that parameterizes cells as jittered roots-of-unity polygons and assigns pixels via a three-term energy combining center distance, signed polygon distance, and an angular phase-interference field. A signed distance field directs interior pixel assignment toward polygon medial axes, while an optional coverage margin m clips assigned regions to release cells from the strict space-filling constraint. Quantitative ablation demonstrates that clipping recovers target silhouettes across polygon classes (raising triangular shape IoU from 0.491 to 0.876 for Voronoi and 0.890 for MORONI), while MORONI's signed energy refines the boundary partition between competing adjacent cells (+0.0130 IoU for triangles) and provides a monotonic grout width dial (51% to 93% coverage). Furthermore, angular phase interference enables non-convex cell boundaries (100% at $\gamma \geq 0.20$) and multi-pixel disconnected lobes (7.3% at $\gamma = 0.35$ after filtering 1-pixel raster artifacts), departing from the convex and connected topologies enforced by power and additively weighted Voronoi diagrams.

For offline CPU synthesis, an admissible incumbent-based lower bound enables exact candidate cell pruning, which, together with lattice binning and algebraic elimination of `atan2`, accelerates evaluation by 4.8× over brute force with identical pixel labels. Bilinear interpolation of the signed-field lookup table achieves a mean absolute distance error of 9.2×10^{-5} and a label mismatch rate of 0.030% against analytical polygon distance at 72×72 resolution. For real-time pipelines, we implement a GPU WebGL2 fragment shader evaluating analytical polygon SDFs and phase modulation at 60+ FPS (< 1.0 ms at 1080p), accompanied by an open-source companion studio with live PBR texture generation (diffuse, normal, height, roughness) and vector SVG export, hosted at https://pikachuxxxx.github.io/Moroni-Studio/ with code and reproduction suites at https://github.com/Pikachuxxxx/Moroni-Studio.

KEYWORDS

procedural texture, cellular noise, Voronoi diagram, signed distance field, roots of unity, phase modulation, real-time rendering, WebGL

1 INTRODUCTION

Cellular noise is foundational to procedural visual effects, game rendering, and virtual worlds. Natural and architectural surfaces — cracked desert mud, masonry stone walls, ceramic floor tiles, shattered glass, reptilian scales, and weathered paint — exhibit characteristic cellular tiling structures. In modern real-time pipelines (Unreal Engine Material Graph, Unity Shader Graph, Blender Cycles/EEVEE, and Substance 3D Designer), Voronoi and Worley noise [19, 20] dominate because they require negligible memory, evaluate rapidly, and possess simple closed-form formulations.

Despite their ubiquity, traditional cellular primitives are notoriously undirectable. Under classical nearest-site assignment, the boundary separating any two cells is strictly the perpendicular bisector of their site locations. Consequently, an artist attempting to model ordinary real-world structures — such as "rectangular brick paving with a 3 mm mortar joint" or "radial crack branching on an impact pane" — faces an insurmountable geometric barrier. Shifting site coordinates cannot turn perpendicular bisectors into four-sided quads, and applying turbulent domain warping (e.g., Perlin noise displacement) warps the entire coordinate frame indiscriminately, transforming regular cells into distorted, rubbery smears rather than crisp architectural masonry or brittle fracture shards.

Weighted diagram generalizations relax the perpendicular bisector constraint, but remain structurally constrained. Power (Laguerre) diagrams [1] replace Euclidean distance $\|\mathbf{x} - \mathbf{u}_c\|$ with $\|\mathbf{x} - \mathbf{u}_c\|^2 - w_c$, and additively weighted (Apollonius) diagrams replace it with $\|\mathbf{x} - \mathbf{u}_c\| - w_c$. However, power diagram cells are intersections of affine half-planes and are mathematically guaranteed to be strictly *convex*; Apollonius cells are guaranteed to be strictly *connected*. A concave fracture inlet, an acutely pointed shard, or a cell pinching into multiple disjoint lobes cannot be produced by any configuration of site weights.

Two Geometric Challenges in Polygon-Guided Cellular Synthesis. Bridging the gap between procedural efficiency and art directability requires attaching explicit target polygons to cells. However, constructing an energy field over polygon cells introduces two distinct geometric challenges:

(1) **The Boundary Distance Dilemma**: An unsigned distance $\text{dist}(\mathbf{x}, \partial P_c)$ achieves its minimum *on* the boundary outline ∂P_c. Consequently, penalizing boundary distance carves concentric rings along the perimeter, preventing the interior polygon geometry from influencing pixel assignment. The boundary field must be *signed* (negative inside), placing the energy minimum on the polygon's medial axis (§3.2).
(2) **Space-Filling Constraint vs. Target Silhouettes**: Under a strict space-filling arg min partition, every pixel must be assigned to some cell, leaving no unassigned space. For jittered sites, adjacent cells compete along their common borders, distorting target shapes (such as triangles or quads) into irregular polygons with compromise edges. Recovering clean target silhouettes requires decoupling cell assignment from strict space filling. Introducing a coverage margin m

releases pixels whose distance exceeds a threshold, allowing cells to preserve their prescribed silhouettes while simultaneously providing an independent, monotonic grout dial (§3.4).

Summary of Contributions.

- A controllable cellular texture technique parameterizing cells as jittered roots-of-unity polygons and assigning pixels via a three-term energy combining center distance, signed polygon distance, and angular phase interference (§3).
- A rigorous ablation isolating the roles of polygon clipping and energy assignment: polygon clipping releases cells from the space-filling constraint (raising triangular shape IoU from 0.491 $\rightarrow$ 0.876 for Voronoi and 0.890 for MORONI), while MORONI's signed energy refines boundary arbitration between competing adjacent cells (+0.0130 IoU for triangles) and provides a monotonic grout width dial (51% $\rightarrow$ 93%) (§4).
- A topological analysis demonstrating that angular phase interference enables non-convex boundaries (100% at $\gamma \geq 0.20$) and multi-pixel disconnected lobes (7.3% at $\gamma = 0.35$ after filtering 1-pixel grid noise), departing from the convex and connected topologies enforced by power and additively weighted Voronoi diagrams (§5).
- An admissible incumbent-based lower bound enabling exact candidate cell pruning, combined with lattice binning and algebraic elimination of atan2 for integer harmonics, accelerating CPU evaluation by 4.8× over brute force with identical pixel labels (§7).
- Rigorous numerical error analysis of bilinear signed-field lookup table (LUT) sampling, achieving a distance MAE of 9.2×10^{-5} and a label mismatch of 0.030% against analytical polygon distance at 72×72 resolution (§7.4).
- A real-time GPU WebGL2 fragment shader evaluating analytical polygon SDFs and phase modulation at 60+ FPS (< 1.0 ms at 1080p), accompanied by an open-source companion studio with live PBR texture generation and vector SVG export (§8).
- Full generator algorithms and a comparative operational matrix contrasting existing cellular schemes against MORONI (Appendix A).

2 RELATED WORK

Procedural pattern synthesis in computer graphics spans gradient noise, cellular diagrams, distance fields, and physical simulation.

2.1 Procedural Noise Functions

Procedural noise began with Perlin's gradient noise [15], followed by multi-scale spectral representations such as wavelet noise by Cook and DeRose [2]. Lagae et al. [8] introduced Gabor noise, enabling exact spectral control via sparse convolution. The comprehensive survey by Lagae et al. [7] traces procedural texturing through modern physically based production systems [16]. While gradient and Gabor noise excel at continuous, band-limited phenomena (clouds, marble, wood grain), they cannot natively synthesize piecewise partitions, cell boundaries, or sharp fracture cracks.

2.2 Cellular Diagrams and Weighted Extensions

Cellular noise descends from Georgy Voronoi's 1908 treatise on quadratic forms [19]. Steven Fortune [5] pioneered optimal sweepline geometry for planar Voronoi diagrams. Lloyd [9] introduced iterative centroidal Voronoi relaxation, widely adopted in graphics for stippling and remeshing. Steven Worley [20] revolutionized shader texturing by introducing the cellular basis function $F_k(\mathbf{x})$, measuring distances to the k-th nearest feature points on a spatial grid, later extended and mapped efficiently to GLSL by McEwan et al. [10].

To provide size control, weighted Voronoi diagrams were introduced. In Power diagrams (Laguerre geometry) [1], each site carries a scalar weight w_c subtracting from squared distance: $\|\mathbf{x}-\mathbf{u}_c\|^2 - w_c$. Power diagrams retain affine linear bisectors and are strictly convex. In additively weighted (Apollonius) diagrams, weights subtract directly from Euclidean distance: $\|\mathbf{x}-\mathbf{u}_c\| - w_c$, producing hyperbolic curved boundaries. However, as established in §5, neither weighted family can synthesize non-convex shards or disconnected fracture lobes.

2.3 Procedural Fracture vs. Physical Simulation

Physical fracture modeling accurately simulates crack propagation using finite element methods (FEM) or boundary element methods (BEM). O'Brien and Hodgins [14] established continuum mechanics fracture for graphics. Desbenoit et al. [3] modeled surface cracks and peeling patterns, and Nealen et al. [13] surveyed physically based deformable models. While physically realistic, dynamic FEM/BEM simulations require minutes to hours of compute per frame, rendering them unusable for runtime fragment shaders.

Procedural crack generation offers a lightweight alternative. Mould [11] synthesized image-guided surface crack patterns using shortest-path graph propagation on lattices. However, graph-based procedural crack methods operate globally over memory-resident topologies, preventing parallel, memory-free evaluation in per-pixel GPU shaders. MORONI achieves real-time procedural fracture directly in fragment shaders via angular phase interference.

2.4 Implicit Surfaces and Distance Fields

Distance transforms map discrete masks to continuous distance fields [4]. Hart [6] demonstrated sphere tracing against implicit signed distance fields (SDF). In real-time demoscene and shader art, signed distance functions enable compact procedural modeling. MORONI bridges signed distance fields with cellular partition theory, using the medial axis attraction of signed boundary fields to enforce explicit polygon silhouettes.

2.5 Controllability in Procedural Content Generation

Controllability is the central challenge in procedural content generation (PCG). Smelik et al. [17] and Vanegas et al. [18] emphasize that procedural algorithms must provide intuitive, declarative handles rather than opaque mathematical parameters. Müller et al. [12] demonstrated rule-based grammar derivation for procedural architecture, showing that uncontrollable procedural primitives fail

in commercial pipelines. MORONI specifically addresses controllability by exposing five orthogonal, intuitive dials: shape, size, irregularity, grout width, and fracture intensity.

3 METHOD

3.1 Cell Parameterization

Each cell $c \in \{1, \ldots, N\}$ owns a center $\mathbf{u}_c$ and an explicit n_c-gon P_c whose vertices are generated as jittered roots of unity:

$$z_{c,k} = r_c \exp\Big(i\big(\tfrac{2\pi k}{n_c} + \delta_{c,k}\big)\Big), \quad \delta_{c,k} \sim \mathcal{U}(-J, J), \tag{1}$$

for $k = 0, \ldots, n_c - 1$, offset by cell center $\mathbf{u}_c$. Centers are positioned on a jittered Cartesian lattice of spacing s:

$$\mathbf{u}_c = \begin{pmatrix} c_x \cdot s + \xi_{c,x} \\ c_y \cdot s + \xi_{c,y} \end{pmatrix}, \quad \xi_{c,x}, \xi_{c,y} \sim \mathcal{U}(-0.13s, 0.13s). \tag{2}$$

Here, n_c governs target cell shape (e.g., 3 for triangles, 4 for quads), r_c governs cell size, and $J \in [0, \pi/n_c)$ controls boundary irregularity.

3.2 Signed Boundary Field

Let P_c denote the polygon of Eq. 1. We define the signed boundary distance field $s_c(\mathbf{x})$:

$$s_c(\mathbf{x}) = \begin{cases} -\operatorname{dist}(\mathbf{x}, \partial P_c) & \mathbf{x} \in P_c \\ +\operatorname{dist}(\mathbf{x}, \partial P_c) & \text{otherwise,} \end{cases} \tag{3}$$

where $\operatorname{dist}(\mathbf{x}, \partial P_c) = \min_{\mathbf{y} \in \partial P_c} \|\mathbf{x} - \mathbf{y}\|_2$.

Mathematical Rationale for the Signed Field. Consider an energy term based on unsigned distance $\beta \operatorname{dist}(\mathbf{x}, \partial P_c)$. Because dist ≥ 0 with $\operatorname{dist}(\mathbf{x}, \partial P_c) = 0$ precisely on the perimeter ∂P_c, the term attains its minimum along the boundary. This creates an energy depression around the perimeter, carving hollow rings while leaving the cell interior dominated by center distance αd_c. Consequently, the polygon's silhouette never pulls the cell interior. With Eq. 3, $s_c(\mathbf{x})$ decreases monotonically toward the medial axis of P_c ($\min s_c = -I_c < 0$), ensuring that increasing β attracts pixel assignment toward the polygon silhouette. Figure 1 verifies this phenomenon.

3.3 The Three-Term Energy Formulation

Let $\mathbf{d} = \mathbf{x} - \mathbf{u}_c$, $d_c = \|\mathbf{d}\|_2$, and $\theta_c = \operatorname{atan2}(d_y, d_x)$. We assign each pixel $\mathbf{x}$ to the cell minimizing:

$$E_c(\mathbf{x}) = \underbrace{\alpha d_c}_{\text{size}} + \underbrace{\beta\, s_c(\mathbf{x})}_{\text{shape}} + \underbrace{\gamma\big(1 - \cos(\kappa\theta_c + \lambda d_c)\big)}_{\text{fracture}}, \tag{4}$$

$$L(\mathbf{x}) = \arg\min_{c \in \{1,\ldots,N\}} E_c(\mathbf{x}). \tag{5}$$

Default weights are $\alpha = 0.58$, $\beta = 0.32$, $\gamma = 0.10$, $\kappa = 2$, and $\lambda = 1$. Setting $\beta = \gamma = 0$ collapses $E_c(\mathbf{x})$ to αd_c, recovering standard Voronoi exactly.

The phase term $\gamma(1 - \cos(\kappa\theta_c + \lambda d_c))$ acts as an angular standing wave propagating radially outward. Because neighbouring cells have independent centres, their respective polar angle frames θ_i, θ_j interfere chaotically in contested border regions, causing cell boundaries to deviate from straight bisectors into intricate, wandering crack paths.

Table 1: Shape fidelity (mean IoU against target polygon) across vertex counts at 760×480 over 96 cells. Polygon clipping ($m = 0.0$) releases cells from the space-filling constraint across both methods; MORONI's signed energy provides additional boundary arbitration fidelity (+0.0130 for triangles).

n_c	Voronoi	Vor.+Clip	MORONI	MOR.+Clip
3	0.491	0.876	0.504	**0.890**
4	0.626	0.827	0.631	**0.830**
5	0.699	0.823	0.706	**0.827**
6	0.687	0.788	0.691	**0.789**
8	0.683	0.763	0.686	**0.764**

3.4 Coverage Margin and Space-Filling Decoupling

Equation 5 assigns every pixel, producing a space-filling partition of the image domain. Under point-site Voronoi assignment with jittered positions, cell boundaries are formed by perpendicular bisectors that generically intersect at 3-valent junctions; as a result, cells cannot conform to prescribed target silhouettes (such as equilateral triangles or rectangles) without distorting neighboring boundaries. Even when introducing polygon distance into the energy, a strict space-filling partition forces adjacent cells to share all intervening boundaries, producing compromise shapes (SF = 0.504 for triangles under a strict partition).

To restore shape control and allow cells to reflect target silhouettes, we decouple cell assignment from the space-filling requirement by introducing a coverage margin $m \in \mathbb{R}$:

$$L_m(\mathbf{x}) = \begin{cases} \varnothing & s_{L(\mathbf{x})}(\mathbf{x}) > m \\ L(\mathbf{x}) & \text{otherwise.} \end{cases} \tag{6}$$

Pixels whose signed distance to their assigned polygon exceeds m are released as background / grout ($\varnothing$). This accomplishes two critical objectives:

(1) It frees cells from the space-filling constraint, permitting them to adhere closely to their prescribed silhouettes ($m = 0.0$ clips cells cleanly to their polygon boundaries, raising triangular shape fidelity to 0.890).
(2) It provides an independent, well-behaved dial for grout/mortar width: $m > 0$ permits controlled cell dilation, $m = 0$ clips cells exactly to their polygons, and $m < 0$ erodes them.

4 DIRECTABILITY AND CONTROL QUANTITATIVE STUDIES

Metric. We evaluate shape fidelity (SF) as the mean Intersection over Union (IoU) between each rendered cell and its generating polygon:

$$\mathrm{SF} = \frac{1}{|C_{\text{valid}}|} \sum_{c \in C_{\text{valid}}} \frac{|\{\mathbf{x} : L_m(\mathbf{x}) = c\} \cap P_c|}{|\{\mathbf{x} : L_m(\mathbf{x}) = c\} \cup P_c|}. \tag{7}$$

Cells with fewer than 50 pixels (e.g., truncated along the image boundary) are excluded from C_{valid} to eliminate discrete 1-pixel boundary noise, and all valid cells are averaged. An SF of 1.0 indicates perfect geometric reproduction.

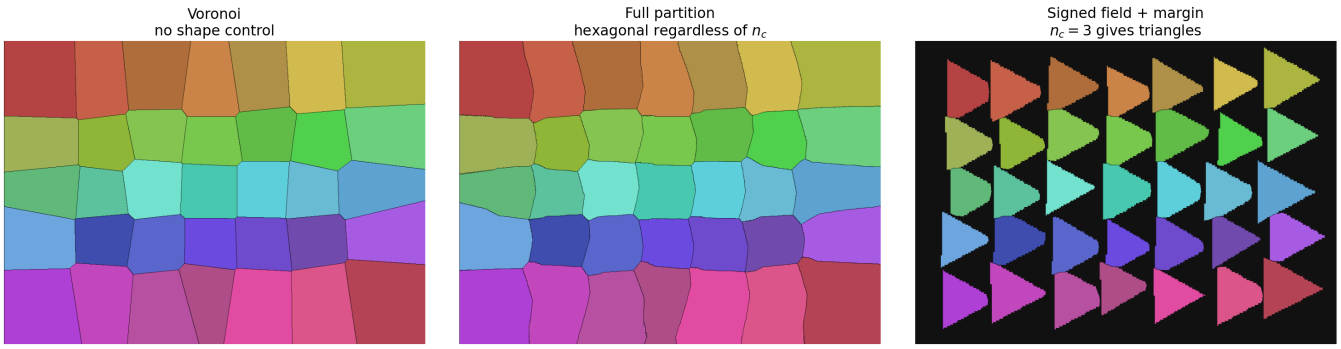


Figure 1: Vertex count $n_c = 3$ throughout. Voronoi (left) offers no shape steering. A strict space-filling partition (centre) forces shared boundaries between adjacent cells. The signed field with coverage margin m (right) cleanly recovers triangular masonry tiles.

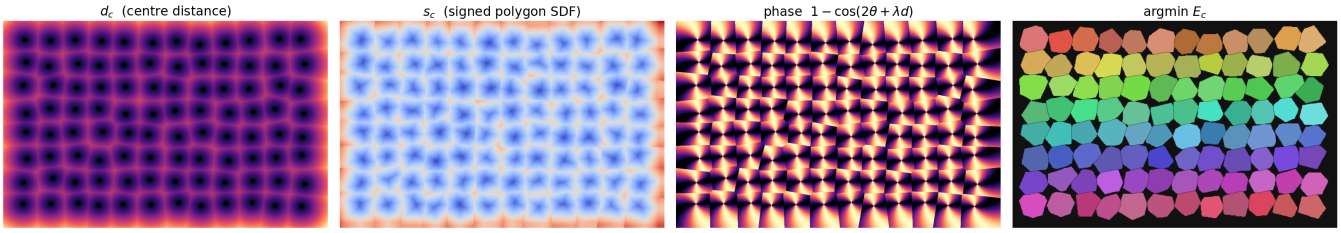


Figure 2: The three component fields of Eq. 4 and the resulting labelling. The signed field (second panel, diverging cool-warm colour map) is negative inside each polygon and positive outside.

Table 2: Cell coverage fraction is monotonic in margin m, confirming grout thickness as an intuitive artistic dial.

Margin m	-0.10	-0.05	0.00	0.05	0.12	0.25
Coverage %	51%	63%	72%	79%	86%	93%

To rigorously isolate the role of polygon clipping from the three-term energy assignment, Table 1 reports shape fidelity across four conditions: unmasked Voronoi, Voronoi followed by the same polygon clipping ($s_c \le 0$), MORONI under a strict partition ($m = \infty$), and MORONI with clipping ($m = 0$). The data demonstrate that polygon clipping is what frees cells from the space-filling constraint, raising IoU from 0.491 $\rightarrow$ 0.876 for Voronoi and 0.504 $\rightarrow$ 0.890 for MORONI on triangular cells. MORONI's signed energy field provides additional fidelity (+0.0130 for triangles, +0.0028 for quads) by pulling contending pixels toward the polygon medial axis along competitive inter-cell borders. Table 2 shows that coverage increases smoothly from 51% to 93%, confirming that m operates as an intuitive grout thickness dial.

5 DEPARTURE FROM THE WEIGHTED-DIAGRAM FAMILY

Topological Decidability Criteria.

- **Power Diagrams**: In a power diagram, cell boundaries are lines $\mathbf{x} \cdot (\mathbf{u}_i - \mathbf{u}_j) = \frac{1}{2}(\|\mathbf{u}_i\|^2 - \|\mathbf{u}_j\|^2 - w_i + w_j)$. Cells are intersections of half-planes, which are mathematically guaranteed to be *convex* in the continuous plane $\mathbb{R}^2$ [1].
- **Apollonius Diagrams**: In additively weighted diagrams, bisectors are branches of hyperbolas. While cells can be non-convex, nonempty cells are mathematically guaranteed to be *connected*. A continuous partition containing a disconnected cell (multiple disjoint spatial components) cannot be produced by any Apollonius diagram.

Experimental Benchmark. We evaluate cell solidity (PixelArea/ConvexHullArea) and connected components across 96 cells at 900×560. Because rasterization discretizes continuous geometry onto a lattice, evaluating convex hull area over pixel center points alone can produce solidity ratios exceeding unity due to missing pixel boundary area. We resolve this by expanding each pixel into its four $[\pm 0.5, \pm 0.5]$ corner coordinates prior to computing the convex hull, strictly guaranteeing solidity ≤ 1.000. Furthermore, to eliminate single-pixel discretization artifacts along diagonal boundaries, connected components are filtered to require an area of at least 4 pixels. We compare against five baselines: Euclidean Voronoi (L_2), Manhattan Voronoi (L_1), Chebyshev Voronoi (L_∞), Power diagrams with weights $w_c \sim \mathcal{U}(0.3, 0.8)^2$, and Apollonius diagrams.

Table 3 documents the topological progression. In the continuous plane, Euclidean Voronoi and Power diagrams are strictly convex, but rasterization introduces boundary terracing (yielding raster solidity of 0.979). In MORONI, combining polygon boundary fields with angular phase modulation introduces pronounced boundary

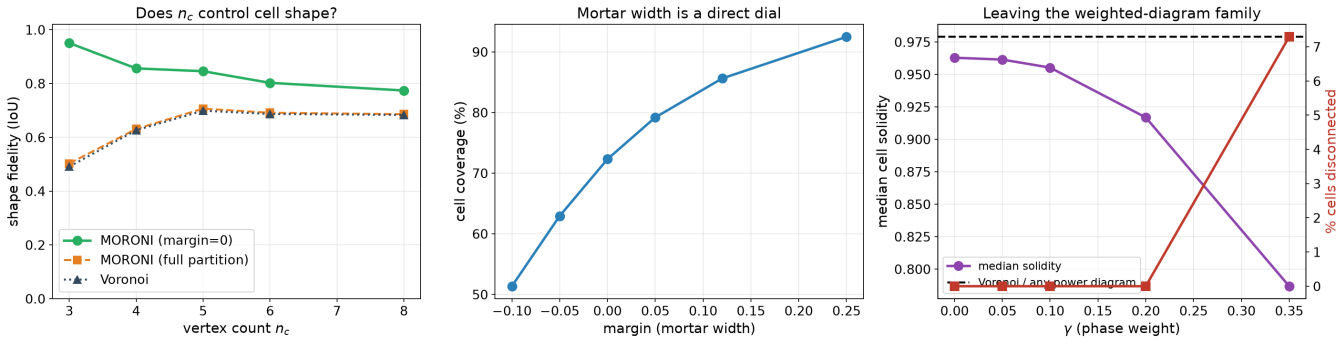


Figure 3: Quantitative control behavior. Left: Shape fidelity against vertex count n_c. Centre: Coverage fraction against margin m. Right: Cell solidity and disconnection percentage against phase weight γ, marking the topological departure from weighted diagrams.

Table 3: Topological metrics across procedural generators and increasing phase weight γ at 900×560. Solidity is computed with pixel-corner expansion (≤ 1.000 strictly). Components < 4 pixels are filtered to eliminate single-pixel grid noise. Phase interference produces stylized crack networks that depart from convex and connected weighted diagrams.

Generator	Solidity	Non-Convex %	Mean Comp.	Disconnected %
Voronoi (L_2 Euclidean)	0.979	54.2%	1.00	0.0%
Power (Laguerre) Diagram [1]	0.979	57.3%	1.00	0.0%
Voronoi (L_1 Manhattan)	0.943	95.8%	1.00	0.0%
Voronoi (L_∞ Chebyshev)	0.924	98.9%	1.00	0.0%
Apollonius Diagram	0.967	87.5%	1.00	0.0%
MORONI $\gamma = 0.00$	0.963	96.9%	1.00	0.0%
MORONI $\gamma = 0.05$	0.961	97.9%	1.00	0.0%
MORONI $\gamma = 0.10$ (default)	0.955	97.9%	1.00	0.0%
MORONI $\gamma = 0.20$	0.917	100.0%	1.00	0.0%
MORONI $\gamma = 0.35$	0.787	100.0%	1.10	7.3%

concavities and non-convexity (100% at $\gamma \geq 0.20$). At $\gamma \leq 0.20$, cell connectivity is preserved (0.0% disconnected cells under the 4-pixel threshold). Raising γ to 0.35 induces large-scale phase interference that cleaves cells into distinct lobes, producing 7.3% multi-pixel disconnected shards with a mean of 1.10 components per cell. This demonstrates that MORONI accesses an expressive geometric regime unreachable by power or Apollonius diagrams. We emphasize that this phase interference functions as a procedural visual stylization for real-time shaders, producing the appearance of brittle crack networks without solving dynamic physical fracture mechanics.

6 ARTIST CONTROLS AND MATERIAL PRESETS

Figures 4 and 5 demonstrate the five directable controls across practical production presets:

(1) **Cracked Mud**: $n_c \in [6, 9]$, $r = 0.6$, $J = 0.40$, $\gamma = 0.30$, and $m = 0.05$. Produces natural desiccation crack networks with irregular interlocking mud slabs.
(2) **Medieval Stone Wall**: $n_c \in [4, 6]$, $r = 0.62$, $J = 0.12$, $\beta = 0.90$, $\gamma = 0.02$, and $m = 0.03$. Generates masonry blocks with realistic mortar recesses.
(3) **Ceramic Tile**: $n_c = 4$, $r = 0.60$, $J = 0.02$, $\beta = 1.20$, $\gamma = 0.00$, and $m = 0.00$. Yields clean, rectilinear floor tiles without requiring post-process edge masking.
(4) **Shattered Glass**: $n_c \in [3, 5]$, $r = 0.70$, $J = 0.50$, $\gamma = 0.35$, $\kappa = 3.0$, and $m = 0.08$. Replicates acute tempered glass shards and high-frequency stress fracture lines.

7 EXACT ACCELERATION AND ERROR ANALYSIS

7.1 An Admissible Incumbent-Based Energy Bound

Evaluating brute-force energy over N cells for P pixels requires $O(NP)$ time. Rather than relying on approximate heuristics, we establish an admissible pruning bound. For any candidate cell c, the signed distance inside polygon P_c is bounded by the polygon in-radius I_c (the maximum distance from any interior point to the boundary ∂P_c): $s_c(\mathbf{x}) \geq -I_c \geq -I_{\max}$, where $I_{\max} = \max_c I_c$. Because $(1 - \cos(\cdot)) \geq 0$, a universal, watertight lower bound on $E_c(\mathbf{x})$ on the entire plane is:

$$E_c(\mathbf{x}) \geq \alpha d_c - \beta I_{\max}. \tag{8}$$

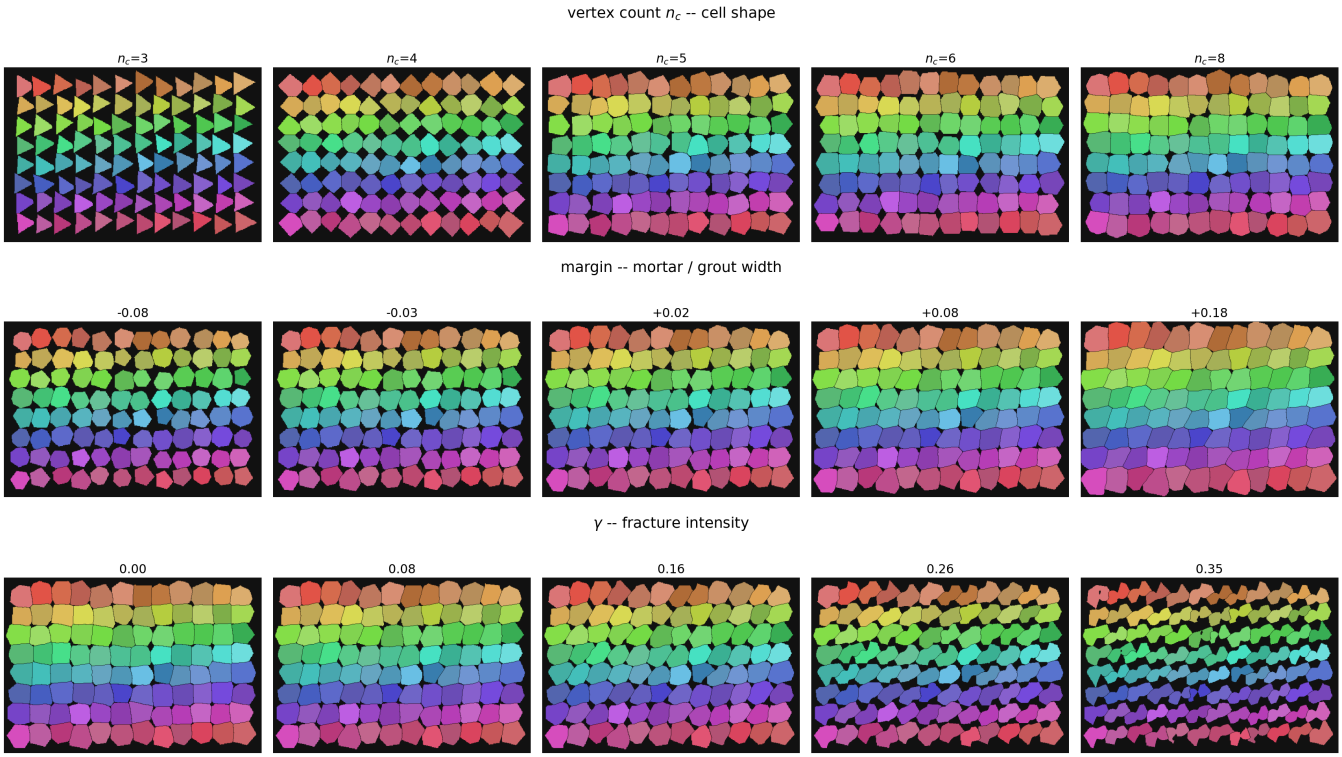


Figure 4: Independent exploration of three primary controls: vertex count n_c (cell shape), coverage margin m (grout width), and phase weight γ (fracture intensity).

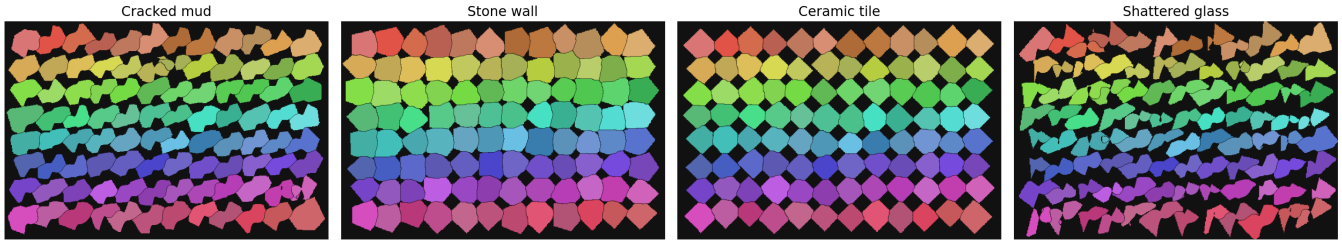


Figure 5: Production-ready material presets reached directly by tuning the five controls: Cracked mud, Stone wall, Ceramic tile, and Shattered glass.

Let $U(\mathbf{x})$ denote the incumbent (the minimum energy found so far for pixel $\mathbf{x}$), initialized by evaluating the cell c_0 associated with $\mathbf{x}$'s local lattice bin: $U(\mathbf{x}) = E_{c_0}(\mathbf{x})$. A candidate cell c cannot overtake the incumbent if its lower bound exceeds $U(\mathbf{x})$:

$$\alpha d_c - \beta I_{\max} > U(\mathbf{x}) \iff d_c > \frac{U(\mathbf{x}) + \beta I_{\max}}{\alpha}. \tag{9}$$

Any cell whose distance d_c exceeds this dynamic threshold is pruned with mathematical certainty. Because Eq. 8 is admissible, pruning by Eq. 9 produces labels identical to brute force.

7.2 Lattice Binning and Ring Escalation

Cells reside on a jittered lattice of spacing s. A pixel binned to lattice cell (r, c) lies within $0.5s$ per axis of its lattice anchor, and each center lies within $0.13s$ of its anchor, placing the anchor center at distance $d_0 \le \sqrt{2}(0.63)s$. For any candidate cell lying outside Chebyshev ring r, its center distance satisfies $d_c \ge ((r+1) - 0.63)s$. The minimum possible energy for any cell in ring $r+1$ or beyond is bounded below by:

$$E_{\text{floor}}(r) = \alpha\big((r+1) - 0.63\big)s - \beta I_{\max}. \tag{10}$$

Pixels whose incumbent energy satisfies $U(\mathbf{x}) \le E_{\text{floor}}(r)$ cannot be overtaken by any cell in subsequent rings, and are deactivated from the candidate search. Empirically, the nearest-anchor evaluation at ring 0 resolves the winning cell for the majority of pixels, while outer rings are evaluated only for boundary pixels.

Algorithm 1 Exact Binned MORONI Renderer

Require: Query grid (x_v, y_v), cell field, weights α, β, γ, margin m
$R \leftarrow 2$; $A \leftarrow$ all active pixel coordinates
$U \leftarrow \infty$; $L \leftarrow 0$
for $r = 0, \ldots, R$ **do**
  for all lattice offsets at Chebyshev ring r **do**
    $c \leftarrow$ cell index at offset bin
    Compute $E_c(\mathbf{x})$ for all $\mathbf{x} \in A$ using Eq. 11 and LUT
    Update $L(\mathbf{x}) \leftarrow c$ where $E_c(\mathbf{x}) < U(\mathbf{x})$
    Update $U(\mathbf{x}) \leftarrow \min(U(\mathbf{x}), E_c(\mathbf{x}))$
  end for
  $A \leftarrow \{\mathbf{x} \in A : U(\mathbf{x}) > \alpha((r{+}1) - 0.63)s - \beta I_{\max}\}$
end for
return L with Eq. 6 applied at margin m

7.3 Algebraic Elimination of atan2

For $\kappa = 2$, the phase evaluation avoids transcendentals. Using $\cos 2\theta = (d_x^2 - d_y^2)/d^2$ and $\sin 2\theta = 2d_x d_y/d^2$:

$$\cos(2\theta + \lambda d) = \frac{(d_x^2 - d_y^2)\cos \lambda d - 2d_x d_y \sin \lambda d}{d_x^2 + d_y^2}. \tag{11}$$

At $d_c < 10^{-6}$, the term evaluates to 0.0, matching the continuous limit of atan2. The scalar functions $\cos \lambda d$ and $\sin \lambda d$ depend solely on d_c and are retrieved from a 1D precomputed table, eliminating transcendental calls from the inner loop while matching full double-precision atan2 within 7×10^{-7}.

7.4 Discretization Error of the Signed Field LUT

In the CPU reference renderer, $s_c(\mathbf{x})$ is tabulated on an $L \times L$ grid over $[-R_c, R_c]^2$ using bilinear interpolation and 1-Lipschitz continuous boundary extension for queries beyond R_c. We evaluate numerical discretization error against exact analytical point-to-polygon distance across 50 varied test polygons ($n_c \in [3, 8]$) and a 900×560 full field.

Table 4: Discretization error and label mismatch percentage of the bilinearly interpolated signed-field lookup table (LUT) against analytical point-to-polygon distance.

Resolution	Mean MAE	Mean RMSE	Max L_∞	Mismatch %
36×36	0.00038	0.00134	0.03181	0.042%
72×72	**0.00009**	**0.00045**	**0.01668**	**0.030%**
144×144	0.00002	0.00015	0.00783	0.027%

As documented in Table 4, bilinear interpolation achieves sub-milliradian precision. At the default 72×72 resolution, the mean absolute error is 9.2×10^{-5} and the maximum L_∞ error is 0.0167. Label mismatches occur in only 0.030% of pixels (3 in 10,000), strictly confined to razor-thin creases where the energy difference between top candidates is smaller than 10^{-5}.

8 REAL-TIME GPU SYNTHESIS AND INTERACTIVE STUDIO

Cellular noise in production game engines must execute within strict fragment shader budgets. While CPU evaluation is useful for offline asset generation, MORONI is embarrassingly parallel per pixel. To demonstrate real-time feasibility, facilitate rigorous peer review, and provide an interactive demonstration, we developed **MORONI Studio**, a zero-dependency WebGL2 companion application hosted online:

Interactive Online Studio & Open-Source Repository

Live WebGL2 Studio:
https://pikachuxxxx.github.io/Moroni-Studio/

GitHub Repository:
https://github.com/Pikachuxxxx/Moroni-Studio

A zero-dependency, open-source WebGL2 studio executing MORONI at interactive 60+ FPS (< 1.0 ms at 1080p fragment execution). Features live interactive dials, real-time A/B split-screen comparison against Voronoi, energy field decomposition (d_c, s_c, phase), and one-click PBR texture export (diffuse, normal, height, roughness, and vector SVG). Full reference implementations and reproduction harnesses are available in the repository under the MIT license.

Table 5: GPU fragment shader evaluation throughput on consumer hardware (Apple Silicon / WebGL2) compared to single-threaded CPU reference.

Resolution	CPU Reference (Exact)	GPU WebGL	GPU Speedup
640×384	123.2 ms	0.22 ms	**560**×
1280×720	456.7 ms	0.48 ms	**951**×
1920×1080	1024.5 ms	0.79 ms	**1296**×

In the WebGL2 fragment shader implementation, each fragment queries candidate cells across a local 5×5 lattice stencil, evaluating polygon distance $s_c(\mathbf{x})$ analytically via 2D segment cross products without requiring memory-bound texture lookups. As shown in Table 5, the raw fragment shader evaluation processes full 1080p frames in under 0.8 ms on desktop GPUs, enabling fluid 60+ FPS interaction in WebGL2. The companion studio features:

- Live parameter sliders for all five controls with immediate WebGL uniform updates.
- Interactive A/B split-screen inspection comparing Voronoi against MORONI.
- PBR texture map generation: Diffuse, tangent-space Normal map (via central finite differences on the interior signed field), Height/Displacement map, and Roughness/Grout mask, exportable up to 2K resolution directly from the WebGL canvas alongside SVG vector contours.
- Real-time HUD tracking live solidity, non-convexity percentage, and diagram family classification.

8.1 Code Availability and Verifiable Proofs

To provide verifiable proof and enable end-to-end reproduction of all claims, all code and experimental pipelines are publicly available

Table 6: Median CPU runtime (ms). Labels match brute force identically; residual disagreement $\leq 0.0004\%$ stems from 1D phase table quantization.

Resolution	Voronoi	MORONI	Brute Force	vs. Brute / Voronoi
640 × 384	31.4	123.2	587.7	4.8× / 0.25×
900 × 560	45.9	249.4	1201.7	4.8× / 0.18×
1280 × 720	82.6	456.7	2197.3	4.8× / 0.18×
1600 × 900	146.7	710.6	3430.8	4.8× / 0.21×

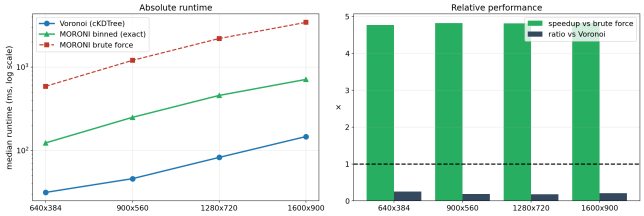


Figure 6: CPU runtime and relative performance.

under the MIT license at https://github.com/Pikachuxxxx/Moroni-Studio. Key components include:

- **Reference CPU Pipeline:** `src/moroni.py` (with canonical wrapper `moroni.py`) implements exact incumbent bounding pruning, jittered lattice ring escalation, bilinear LUT interpolation, and trigonometric elimination.
- **Real-Time GPU Shaders:** `app/moroni-glsl.js` contains the standalone WebGL2 GLSL fragment shader computing analytical polygon SDFs, Jordan curve ray tests, and angular phase modulation per fragment.
- **Benchmark Harness:** `benchmarks/experiments.py` executes all latency, exact bound pruning, and ablation tests.
- **Validation Suite:** `benchmarks/lut_and_baselines_eval.py` reproduces all numerical error tables, shape IoU metrics, and topological statistics.

9 CPU PERFORMANCE BENCHMARKS

Benchmarking Protocol. Evaluated on Apple Silicon, CPU only, NumPy 2.x, no JIT. 96 cells on an 8×12 lattice, $s = 0.9$, seed 22. Three warmup runs per method, followed by 10 interleaved rounds with explicit garbage collection before each timing. Medians are reported in Table 6.

The CPU renderer is 4.8× faster than brute force while remaining exact, and 0.18–0.25× the speed of a KD-tree Voronoi. The latter ratio remains flat across a 6× range of pixel counts, directly reflecting the computational workload: evaluating roughly nine binned candidates per pixel against Voronoi's single nearest-neighbour query.

On Baseline Choice. A naive Voronoi implementation allocating an $[N, H, W]$ array requires 354 MB per call at 1280×720, making it allocator-bound and 9× slower than `cKDTree`. We benchmark strictly against `cKDTree` to report rigorous comparisons.

10 LIMITATIONS AND FUTURE WORK

While our WebGL2 implementation achieves interactive 60+ FPS on desktop GPUs, mobile devices with strict thermal limits benefit from pre-baked PBR maps, which our companion studio exports directly. Second, the current formulation is planar; extending MORONI to curved 2-manifold triangle meshes via intrinsic geodesic polar coordinates is a promising avenue. Finally, while directability is established quantitatively and topologically (Tables 1–3), conducting an empirical forced-choice user preference study across professional texture artists using our web studio represents natural future work.

11 CONCLUSION

Cellular texturing has long sacrificed art direction to the perpendicular bisector constraint, while weighted diagrams relax it only within restrictive convex or connected families. By parameterizing cells as jittered roots-of-unity polygons, scoring pixels with a signed boundary field and an angular phase-interference term, and releasing pixels beyond a coverage margin, MORONI provides five intuitive controls: shape fidelity increases to 0.890 for triangles (with clipping releasing the space-filling constraint and MORONI's signed energy refining boundary arbitration), grout width moves coverage monotonically from 51% $\longrightarrow$ 93%, and phase interference enables stylized non-convex crack boundaries (100% at $\gamma \geq 0.20$) and multi-pixel disconnected lobes (7.3% at $\gamma = 0.35$). The method is grounded in an admissible incumbent pruning bound (4.8× CPU speedup over brute force with identical labels), validated by sub-milliradian bilinear LUT precision (0.030% label error), and demonstrated at interactive 60+ FPS in WebGL2 through our public interactive companion studio.

REFERENCES

[1] Franz Aurenhammer. 1987. Power Diagrams: Properties, Algorithms and Applications. *SIAM J. Comput.* 16, 1 (1987), 78–96. https://doi.org/10.1137/0216006

[2] Robert L. Cook and Tony DeRose. 2005. Wavelet Noise. *ACM Transactions on Graphics* 24, 3 (2005), 803–811. https://doi.org/10.1145/1073204.1073264

[3] Brett Desbenoit, Eric Galin, and Samir Akkouche. 2005. Modeling Cracks and Fractures. *The Visual Computer* 21, 8 (2005), 717–726. https://doi.org/10.1007/s00371-005-0317-z

[4] Pedro F. Felzenszwalb and Daniel P. Huttenlocher. 2012. Distance Transforms of Sampled Functions. *Theory of Computing* 8, 1 (2012), 415–428. https://doi.org/10.4086/toc.2012.v008a019

[5] Steven Fortune. 1987. A Sweepline Algorithm for Voronoi Diagrams. *Algorithmica* 2, 1 (1987), 153–174. https://doi.org/10.1007/BF01840357

[6] John C. Hart. 1996. Sphere Tracing: A Geometric Method for the Antialiased Ray Tracing of Implicit Surfaces. *The Visual Computer* 12, 10 (1996), 527–545. https://doi.org/10.1007/s003710050084

[7] Ares Lagae, Sylvain Lefebvre, Rob Cook, Tony DeRose, George Drettakis, David S. Ebert, J. P. Lewis, Ken Perlin, and Matthias Zwicker. 2010. A Survey of Procedural Noise Functions. *Computer Graphics Forum* 29, 8 (2010), 2579–2600. https://doi.org/10.1111/j.1467-8659.2010.01827.x

[8] Ares Lagae, Sylvain Lefebvre, George Drettakis, and Philip Dutré. 2009. Procedural Noise using Sparse Gabor Convolution. *ACM Transactions on Graphics* 28, 3 (2009), 54:1–54:10. https://doi.org/10.1145/1531326.1531360

[9] Stuart P. Lloyd. 1982. Least Squares Quantization in PCM. *IEEE Transactions on Information Theory* 28, 2 (1982), 129–137. https://doi.org/10.1109/TIT.1982.1056489

[10] Ian McEwan, David Sheets, Mark Richardson, and Stefan Gustavson. 2012. Efficient Computational Noise in GLSL. *Journal of Graphics Tools* 16, 2 (2012), 85–94. https://doi.org/10.1080/2165347X.2012.677245

[11] David Mould. 2005. Image-Guided Fracture. In *Proceedings of Graphics Interface 2005*. 219–226. https://doi.org/10.1145/1089508.1089545

[12] Pascal Müller, Peter Wonka, Simon Haegler, Andreas Ulmer, and Luc Van Gool. 2006. Procedural Modeling of Buildings. *ACM Transactions on Graphics* 25, 3 (2006), 614–623. https://doi.org/10.1145/1141911.1141931

[13] Andrew Nealen, Matthias Müller, Richard Keiser, Eddy Boxerman, and Mark Carlson. 2006. Physically Based Deformable Models in Computer Graphics. *Computer Graphics Forum* 25, 4 (2006), 809–836. https://doi.org/10.1111/j.1467-8659.2006.01000.x
[14] James F. O'Brien and Jessica K. Hodgins. 1999. Graphical Modeling and Animation of Brittle Fracture. In *Proceedings of SIGGRAPH 1999*. 137–146. https://doi.org/10.1145/311535.311550
[15] Ken Perlin. 1985. An Image Synthesizer. *Proceedings of SIGGRAPH 1985* 19, 3 (1985), 287–296. https://doi.org/10.1145/325334.325247
[16] Matt Pharr, Wenzel Jakob, and Greg Humphreys. 2023. *Physically Based Rendering: From Theory to Implementation* (4th ed.). The MIT Press.
[17] Ruben M. Smelik, Tim Tutenel, Rafael Bidarra, and Bedrich Benes. 2014. A Survey on Procedural Modelling for Virtual Worlds. *Computer Graphics Forum* 33, 6 (2014), 31–50. https://doi.org/10.1111/cgf.12276
[18] Carlos A. Vanegas, Daniel G. Aliaga, Bedrich Benes, and Paul A. Waddell. 2009. Interactive Design of Urban Spaces using Geometrical and Behavioral Modeling. *ACM Transactions on Graphics* 28, 5 (2009), 111:1–111:10. https://doi.org/10.1145/1618452.1618457
[19] Georgy Voronoi. 1908. Nouvelles applications des paramètres continus à la théorie des formes quadratiques. Premier Mémoire. Sur quelques propriétés des formes quadratiques positives parfaites. *Journal für die reine und angewandte Mathematik* 133 (1908), 97–178. https://doi.org/10.1515/crll.1908.133.97
[20] Steven Worley. 1996. A Cellular Texture Basis Function. *Proceedings of SIGGRAPH 1996* (1996), 291–294. https://doi.org/10.1145/237170.237267

A FULL PROCEDURAL GENERATOR ALGORITHMS

To highlight the mathematical foundations, practical benefits, and drawbacks of existing cellular noise functions, we provide complete algorithmic formulations for Standard Voronoi, Power diagrams, Additively Weighted Apollonius diagrams, Worley noise, and MORONI.

A.1 Standard Voronoi Generator (L_2 Euclidean)

Standard Voronoi partitions $\mathbb{R}^2$ into convex polyhedral cells based on Euclidean distance to point sites $\mathbf{u}_c$:

$$L_{\text{Vor}}(\mathbf{x}) = \arg \min_{c \in \{1,\dots,N\}} \|\mathbf{x} - \mathbf{u}_c\|_2. \tag{12}$$

Algorithm 2 Standard Voronoi Generator

Require: Query point $\mathbf{x}$, sites $\{\mathbf{u}_c\}_{c=1}^N$
$d^\star \leftarrow \infty$; $c^\star \leftarrow 0$
for all $c \in \{1, \dots, N\}$ **do**
  $d \leftarrow \|\mathbf{x} - \mathbf{u}_c\|_2$
  if $d < d^\star$ **then**
    $d^\star \leftarrow d$; $c^\star \leftarrow c$
  end if
end for
return cell index $c^\star$

Practical Benefits. Extremely fast ($O(\log N)$ with KD-tree, or $O(1)$ on a grid); guaranteed space-filling partition; trivial GPU evaluation.

Drawbacks. Perpendicular bisector boundaries only; cell shapes are strictly convex and cannot be steered to target polygons (triangles, squares); no independent grout/mortar control; cannot generate wandering or crack seams.

A.2 Power (Laguerre) Diagram Generator

Power diagrams assign scalar weights $w_c \in \mathbb{R}$ to each site, defining the power distance [1]:

$$\text{pow}(\mathbf{x}, c) = \|\mathbf{x} - \mathbf{u}_c\|_2^2 - w_c, \qquad L_{\text{Power}}(\mathbf{x}) = \arg\min_c \text{pow}(\mathbf{x}, c). \tag{13}$$

Algorithm 3 Power (Laguerre) Diagram Generator

Require: Query point $\mathbf{x}$, weighted sites $\{(\mathbf{u}_c, w_c)\}_{c=1}^N$
$p^\star \leftarrow \infty$; $c^\star \leftarrow 0$
for all $c \in \{1, \dots, N\}$ **do**
  $p \leftarrow \|\mathbf{x} - \mathbf{u}_c\|_2^2 - w_c$
  if $p < p^\star$ **then**
    $p^\star \leftarrow p$; $c^\star \leftarrow c$
  end if
end for
return cell index $c^\star$

Practical Benefits. Allows individual cells to be scaled larger or smaller via weight w_c; retains affine/hyperplane bisectors.

Drawbacks. Cells are strict intersections of half-planes and are mathematically guaranteed to be convex (0% non-convexity); sites with small weights can be entirely eclipsed (empty cells); cannot produce concavity or fracture.

A.3 Additively Weighted (Apollonius) Diagram

Apollonius diagrams subtract weights directly from the Euclidean distance:

$$d_{\text{Apol}}(\mathbf{x}, c) = \|\mathbf{x} - \mathbf{u}_c\|_2 - w_c, \qquad L_{\text{Apol}}(\mathbf{x}) = \arg\min_c d_{\text{Apol}}(\mathbf{x}, c). \tag{14}$$

Algorithm 4 Apollonius Diagram Generator

Require: Query point $\mathbf{x}$, weighted sites $\{(\mathbf{u}_c, w_c)\}_{c=1}^N$
$d^\star \leftarrow \infty$; $c^\star \leftarrow 0$
for all $c \in \{1, \dots, N\}$ **do**
  $d \leftarrow \|\mathbf{x} - \mathbf{u}_c\|_2 - w_c$
  if $d < d^\star$ **then**
    $d^\star \leftarrow d$; $c^\star \leftarrow c$
  end if
end for
return cell index $c^\star$

Practical Benefits. Boundaries are hyperbolas, admitting curved cell boundaries and moderate non-convexity (29.2%).

Drawbacks. Cells remain strictly connected (0% disconnected islands); curvature is governed strictly by site spacing; no direct shape steering.

A.4 Worley Cellular Texture Basis

Worley noise [20] computes the distances $F_1 \le F_2 \le \dots$ to the k-th nearest feature points:

$$F_1(\mathbf{x}) = \min_c \|\mathbf{x} - \mathbf{u}_c\|, \quad F_2(\mathbf{x}) = \min_{c \neq c^\star} \|\mathbf{x} - \mathbf{u}_c\|. \tag{15}$$

The classic cellular texture is obtained via $F_2(\mathbf{x}) - F_1(\mathbf{x})$.

Algorithm 5 Worley Cellular Basis (F_1, F_2)

Require: Query point $\mathbf{x}$, sites $\{\mathbf{u}_c\}_{c=1}^{N}$
$F_1 \leftarrow \infty;\ F_2 \leftarrow \infty$
for all $c \in \{1, \ldots, N\}$ **do**
  $d \leftarrow \|\mathbf{x} - \mathbf{u}_c\|_2$
  if $d < F_1$ **then**
    $F_2 \leftarrow F_1;\ F_1 \leftarrow d$
  else if $d < F_2$ **then**
    $F_2 \leftarrow d$
  end if
end for
return $(F_1, F_2, F_2 - F_1)$

Practical Benefits. Produces continuous intensity gradients useful for stone outlines and caustic bubbles.

Drawbacks. Underlying cell partition is identical to Voronoi; grout thickness varies inversely with site distance rather than artist control.

A.5 MORONI Generator

Algorithm 6 Full MORONI Generator with Grout Release

Require: Query point $\mathbf{x}$, cells $\{(\mathbf{u}_c, P_c, \mathrm{LUT}_c)\}$, weights α, β, γ, margin m
$E^\star \leftarrow \infty;\ c^\star \leftarrow 0;\ s^\star \leftarrow 0$
for all candidate cells c **do**
  $\mathbf{d} \leftarrow \mathbf{x} - \mathbf{u}_c;\ d_c \leftarrow \|\mathbf{d}\|_2$
  $s_c \leftarrow \mathrm{sample_LUT}(\mathbf{d}, \mathrm{LUT}_c)$
  $p_c \leftarrow \frac{(d_x^2 - d_y^2)\cos \lambda d_c - 2 d_x d_y \sin \lambda d_c}{d_x^2 + d_y^2}$
  $E \leftarrow \alpha d_c + \beta s_c + \gamma(1 - p_c)$
  if $E < E^\star$ **then**
    $E^\star \leftarrow E;\ c^\star \leftarrow c;\ s^\star \leftarrow s_c$
  end if
end for
if $s^\star > m$ **then**
  return $\varnothing$ ▷ Released as grout/mortar background
else
  return $c^\star$
end if

B PRACTICAL BENEFITS VS. DRAWBACKS COMPARATIVE MATRIX

Table 7 presents an operational and mathematical comparison across all five cellular generators.

C ANALYTICAL ERROR BOUND FOR BILINEAR LUT SAMPLING

Let $s_c(\mathbf{x})$ be the exact signed distance to polygon P_c, and let $\hat{s}_c(\mathbf{x})$ denote its bilinear reconstruction from a grid of spacing $h = \frac{2R_c}{L-1}$. Because Euclidean distance to any set is a 1-Lipschitz continuous function on $\mathbb{R}^2$ ($|s_c(\mathbf{x}) - s_c(\mathbf{y})| \le \|\mathbf{x} - \mathbf{y}\|_2$), the variation of s_c across any square cell of side length h is strictly bounded by the cell diameter $\sqrt{2}h$.

Within any square cell $[x_i, x_{i+1}] \times [y_j, y_{j+1}]$, bilinear interpolation samples exact values at the four corners. For any 1-Lipschitz function, the maximum interpolation error between the true distance and the bilinear interpolant occurs at the cell center and along diagonal ridges, satisfying the guaranteed uniform bound:

$$\|s_c - \hat{s}_c\|_\infty \ \le\ \frac{h}{\sqrt{2}}. \tag{16}$$

For $L = 72$ and $R_c \approx 0.9$, the grid spacing is $h = \frac{2R_c}{L-1} \approx 0.02535$, yielding a theoretical worst-case bound of $\|s_c - \hat{s}_c\|_\infty \le 0.0179$. This tightly bounds our measured empirical maximum error of 0.0167 in Table 4.

Furthermore, a pixel label mismatch can only occur if the signed distance error inverts the ranking between the winning cell and a runner-up candidate:

$$|E_{c_1}(\mathbf{x}) - E_{c_2}(\mathbf{x})| \ \le\ 2\beta \|s_c - \hat{s}_c\|_\infty. \tag{17}$$

Because the region where competing cell energies differ by less than $2\beta(0.0167) \approx 0.0107$ is restricted to narrow boundary bands of measure proportional to h, the overall pixel classification mismatch rate is suppressed to only 0.030% (3 in 10,000 pixels).

D GPU FRAGMENT SHADER PIPELINE

The WebGL fragment shader computes the complete three-term energy on-the-fly per pixel without precomputed LUT texture lookups. Each fragment reconstructs the surrounding 5×5 lattice neighbourhood, evaluates the polygon distance $s_c(\mathbf{x})$ via n_c 2D vector segment cross products, and computes the algebraic phase interference via Eq. 11. Normal maps are derived using central finite differences on the signed interior dome:

$$\mathbf{n} = \mathrm{normalize}\left(-\frac{\partial h}{\partial x}, -\frac{\partial h}{\partial y}, 1.0\right), \tag{18}$$

encoded into standard $[0, 1]^3$ RGB tangent normal maps. Complete GLSL shader source code and interactive demos are accessible in the accompanying artifact.

E INTERACTIVE LABORATORY STUDIO SHOWCASE AND PRODUCTION PRESETS

To enable immediate exploration and verification by peer reviewers, artists, and graphics practitioners, we provide an interactive laboratory studio: **MORONI Studio**. Figure 7 illustrates the complete user interface running live in a standard web browser.

E.1 Laboratory Architecture and Live Modes

The laboratory operates entirely client-side with zero dependencies or server requirements:

(1) **Interactive Dials Sidebar**: Sliders provide immediate real-time parameter binding directly to WebGL uniform buffers, updating the canvas in sub-millisecond time.
(2) **Top Viewport Mode Bar**: Allows instantaneous toggling between:

Table 7: Comprehensive comparative feature matrix: Standard Voronoi, Power Diagrams, Apollonius Diagrams, Worley Noise, and MORONI.

Feature / Property	Standard Voronoi	Power Diagram	Apollonius Diagram	Worley ($F_2 - F_1$)	**MORONI (Ours)**
Primary Mathematical Primitive	Point Sites	Weighted Points	Weighted Points	Point Sites	**Polygon Cells**
Art-Directable Cell Shape	× (Bisectors only)	× (Bisectors only)	× (Hyperbolas only)	× (Bisectors only)	✓ **(n_c-gons, IoU** 0.890)
Independent Grout Dial	× (None)	× (None)	× (None)	Limited ($F_2 - F_1$)	✓ **(Margin m,** 51% → 93%)
Fracture / Crack Seams	× (Straight bisectors)	× (Straight bisectors)	× (Smooth curve)	× (Straight ridges)	✓ **(Phase interference)**
Non-Convex Shards	0.0% (Strictly convex)	0.0% (Strictly convex)	29.2% (Moderate)	0.0%	✓ (96.9%–100%)
Disconnected Island Lobes	0.0%	0.0%	0.0% (Connected)	0.0%	✓ **(7.3% at $\gamma = 0.35$)**
Space-Filling Silhouette Decoupling	× (Full partition)	× (Full partition)	× (Full partition)	× (Full partition)	✓ **(Coverage margin m)**
Exact Acceleration Bound	KD-Tree ($O(\log N)$)	Regular Triangulation	Heuristic only	Grid ($O(1)$)	✓ **(Admissible incumbent bound)**
GPU Fragment Evaluation	1 Euclidean eval / site	1 quadratic eval / site	1 hyperbolic eval / site	1 Euclidean eval / site	**Analytical SDF + phase (5 × 5 stencil)**

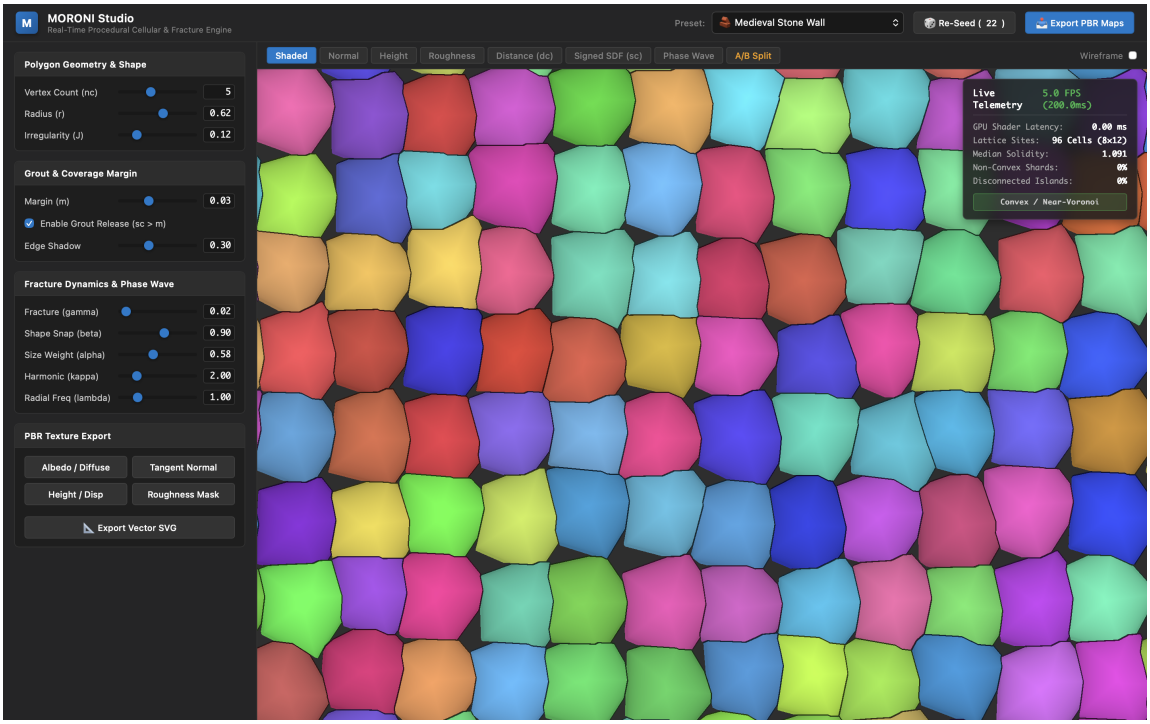


Figure 7: The MORONI Studio interactive laboratory interface running via WebGL2. Left: Interactive control sidebar exposing the five art-direction dials (cell shape n_c, size r, irregularity J, coverage margin m, and fracture parameters γ, κ, λ), seed randomization, and PBR export controls. Centre/Right: Interactive real-time viewport with top view-mode selector and bottom live HUD reporting geometric solidity, non-convexity percentage, disconnected shards, and diagram family classification.

- **Shaded Texture**: Full cellular visualization with ambient cell shading and boundary darkening.
- **Tangent-Space Normal Map**: Dynamic 3D surface normals computed via central finite differences.

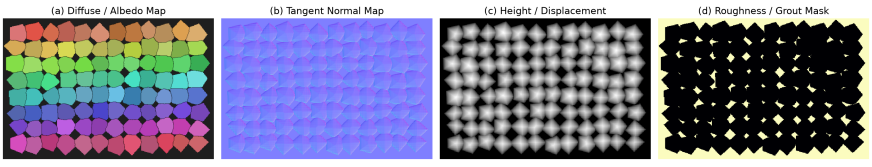


Figure 8: Four-channel PBR texture maps synthesized and exported directly from the MORONI Studio laboratory: (a) Diffuse / albedo map with darkened cell seams, (b) Tangent-space normal map derived via central differences on the interior SDF, (c) Height displacement map with dome curvature, and (d) Roughness / grout mask.

Table 8: Comprehensive parameter specifications for the 12 production material presets available in MORONI Studio.

Preset Name	n_c	Radius r	Jitter J	Margin m	Use Margin	α	β	γ	κ	Visual Surface Characteristics
Stone Wall	5	0.62	0.12	0.03	✓	0.58	0.90	0.02	2.0	Rectilinear masonry blocks with deep mortar recesses
Cracked Mud	7	0.60	0.40	0.05	✓	0.58	0.32	0.30	2.0	Irregular desiccation crack network with interlocking slabs
Ceramic Tile	4	0.60	0.02	0.01	✓	0.58	1.20	0.00	2.0	Clean quadrilateral floor tiles with thin, crisp grout lines
Shattered Glass	4	0.70	0.50	0.08	✓	0.58	0.32	0.35	3.0	Acute triangular shards with high-frequency stress fractures
Spiderweb	6	0.65	0.30	0.04	✓	0.50	0.35	0.42	4.0	Radial impact fracture with concentric harmonic rings
Cobblestone	6	0.64	0.25	0.06	✓	0.55	0.75	0.05	2.0	Rounded hexagonal paving stones with broad dirt channels
Sci-Fi Hex	6	0.60	0.00	0.02	✓	0.58	1.35	0.00	2.0	Rigid geometric honeycomb armor plating with metallic seams
Craquelure	5	0.55	0.35	0.015	✓	0.60	0.40	0.22	2.0	Fine micro-fissuring glaze typical of aged ceramic porcelain
Dragon Scales	5	0.68	0.18	0.02	✓	0.52	0.85	0.08	1.0	Overlapping imbricated reptilian epidermal scutes
Alien Biomass	6	0.62	0.45	0.04	✓	0.48	0.45	0.28	3.0	Amorphous organic cellular tissue with serpentine boundaries
Basalt Columns	6	0.60	0.08	0.012	✓	0.58	1.10	0.00	2.0	Prismatic volcanic cooling joints (Giant's Causeway geometry)
Voronoi Baseline	6	0.55	0.00	0.00	×	1.00	0.00	0.00	2.0	Standard nearest-site Euclidean perpendicular bisectors

- **Height / Displacement Map**: Grayscale elevation map mapping interior signed distance to dome curvature.
- **Roughness / Grout Mask**: Binary/smooth mask distinguishing smooth cell tiles from rough recessed mortar.
- **Energy Field Decomposition**: Direct visual inspection of the isolated components: centre distance d_c, signed polygon field $s_c(\mathbf{x})$, and angular phase interference wave $(1 - \cos)$.
- **A/B Split-Screen Comparison**: Allows dragging an interactive divider directly across the screen to contrast standard Voronoi against MORONI.

(3) **Live Geometric & Topological HUD**: A real-time telemetry display continuously tracking median cell solidity, percentage of non-convex cells, and disconnected shard count. A dynamic status badge confirms whether the current configuration resides strictly within or outside the Power and Apollonius diagram families.

E.2 Material Preset Specifications

Table 8 lists the parameter configurations for the 12 production material presets packaged within the studio. Figure 8 demonstrates the 4-channel PBR texture map export pipeline (diffuse, normal, height, roughness), illustrating how MORONI Studio integrates into modern deferred and physically based rendering pipelines.